

Numerical Method

CSIT — 3rd Semester — 2078 — Answer Sheet
Subject Code: CSC212

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How can Horner's rule be used to evaluate the $f(x)$ and $f'(x)$ of a polynomial at given point? Explain. Write an algorithm and program to calculate a real root of a polynomial using Horner's rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is matrix factorization? How can it be used to solve system of linear equations? Factorize the given matrix A and solve the system of equations $Ax = b$ for given b using L and U matrices. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 11 \\ 3 & 22 & 36 \end{bmatrix}$ and $b = \begin{bmatrix} 12 \\ 32 \\ 28 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is higher order differential equation? How can you solve the higher order differential equation? Explain. Solve the following differential equation for $1 \leq x \leq 2$, taking $h = 0.25$. $d^2y/dx^2 + 3 dy/dx + 5y = 0$, with $y(1) = 1$ and $y'(1) = 2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How the half-interval method can estimate a root of non-linear equation? Find a real root of following equation using half-interval method correct up to two decimal places. $x^2 - e^x - x = 1$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Calculate a real root of the given equation using fixed point iteration correct up to 3 significant figures. $2x^3 - 2x = 5$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is Newton's interpolation? Obtain the divided difference table from the following data set and estimate the $f(x)$ at $x = 2$ and $x = 5$.

x	f(x)
1.0	1.22
1.2	1.71
1.4	2.04
1.6	2.85
1.8	3.6
2.0	4.17
2.2	4.81
2.4	5.23
2.6	5.35
2.8	5.1

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is linear regression? Fit the linear function to the following data.

x	y
1.0	1.21
1.2	1.41
1.4	1.61
1.6	1.82
1.8	2.02
2.0	2.22
2.2	2.42
2.4	2.62
2.6	2.82
2.8	3.02

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What are the problems with polynomial interpolation for large number of data set? How such problems are addressed? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Evaluate the following integration using Romberg integration. $\int_0^1 (\sin^2 x)/x \, dx$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Solve the following set of linear equations using Gauss Jordan method.

$$\begin{aligned} x + 2x + 3x &= 97 \\ x + 5x + 4x &= 338 \\ x + 9x + x &= 272 \\ x + 5x + 4x + 3x &= 23 \end{aligned}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Solve the following differential equation for $1 \leq x \leq 2$, taking $h = 0.25$ using Heun's method. $y'(x) + x^2y = 3x$, with $y(1) = 1$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Consider a metallic plate of size 90 cm by 90 cm. The two adjacent sides of the plate are maintained at temperature of 100°C and remaining two adjacent sides are held at 200°C. Calculate the steady state temperature at interior points assuming a grid size of 30 cm by 30 cm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.